# HELIXCLUSTER PHASE 2 — CONSOLE COMPUTE NODES

## PlayStation 3/4/4 Pro/5/5 Pro Integration Architecture

## Version 1.0 | 2026-05-30

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## 1. EXECUTIVE SUMMARY

HelixCluster Phase 2 extends the distributed computing cluster to include jailbroken PlayStation consoles (PS4, PS4 Pro, PS5, PS5 Pro) as fully integrated worker nodes. **PS3 is excluded** — its Cell BE architecture is obsolete and programming complexity is prohibitive.

### Key Value Proposition

| Metric | PS4 Pro Node | PS5 Node | Desktop PC Equivalent |
|---|---|---|---|
| **Cost (used)** | $150-250 | $400-500 | $600-1000 |
| **GPU TFLOPS** | 4.2 | 10.3 | 4-10 |
| **CPU** | 8c Jaguar @ 2.1GHz | 8c/16t Zen2 @ 3.5GHz | Varies |
| **RAM** | 8GB GDDR5 | 16GB GDDR6 | 16GB DDR4 |
| **GPU $/TFLOP** | ~$59 | ~$49 | ~$100+ |
| **Power** | ~160W | ~200W | ~300W+ |

### Unique Capabilities Consoles Bring

1. **GPU Compute at 1/2 the PC cost** — Discarded gaming hardware repurposed for compute
2. **PS5 Custom I/O Decompressor** — 8-9 GB/s hardware decompression (no PC equivalent)
3. **GDDR5/GDDR6 unified memory** — Higher bandwidth than DDR4 for GPU workloads
4. **Disposable node model** — At $80-250, failed nodes are replaced, not repaired
5. **Community contribution** — Users can donate idle console time (Folding@home model)

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## 2. SUPPORTED CONSOLE HARDWARE MATRIX

### 2.1 Console Tier Classification

```
┌──────────────────────────────────────────────────────────────────┐
│                    CONSOLE TIER CLASSIFICATION                   |
├──────────────┬──────────────────┬──────────────┬─────────────────┤
| TIER         | HARDWARE         | SUITABILITY  | CLUSTER ROLE    |
├──────────────┼──────────────────┼──────────────┼─────────────────┤
| TIER-1       | PS5 / PS5 Pro    | EXCELLENT    | GPU Compute Ace |
| (Premium)    | Zen2 8c/16t      |              | AI Inference Primary |
|              | RDNA2 10-33 TF   |              | High-Perf Batch Jobs |
|              | 16GB GDDR6       |              | PS5 I/O Decompression |
├──────────────┼──────────────────┼──────────────┼─────────────────┤
| TIER-2       | PS4 Pro          | GOOD         | GPU Compute Worker |
| (Standard)   | Jaguar 8c        |              | Batch Job Worker |
```

```
|             | GCN 4.2 TF     |              | Build Farm Node       |
|             | 8GB GDDR5      |              | Video Transcode       |
├─────────────┼────────────────┼──────────────┼───────────────────────┤
| TIER-3      | PS4 Base       | ADEQUATE     | Lightweight Worker    |
| (Basic)     | Jaguar 8c      |              | Cache/Storage Node    |
|             | GCN 1.84 TF    |              | Network Relay         |
|             | 8GB GDDR5      |              | Fallback Compute      |
├─────────────┼────────────────┼──────────────┼───────────────────────┤
| EXCLUDED    | PS3 / PS3 Slim | POOR         | Not Supported         |
|             | Cell BE 7 SPEs |              |                       |
|             | 256MB XDR RAM  |              |                       |
└─────────────┴────────────────┴──────────────┴───────────────────────┘
```

### 2.2 Detailed Hardware Specifications

### PS4 Base (CUH-10xx/11xx series)

```yaml
console:
  model: "PS4 Fat"
  tier: 3
  soc: "AMD Liverpool"

cpu:
  architecture: "AMD Jaguar x86-64"
  cores: 8
  threads: 8
  clock: "1.6 GHz base (up to 1.75 boost)"
  features: ["AES-NI", "AVX", "SSE4.1", "SSE4.2"]
  # NOTE: NO AVX2

gpu:
  architecture: "AMD GCN 2.0 (Liverpool)"
  compute_units: 18
  tflpos_fp32: 1.84
  memory: "Shared with system"

memory:
  type: "GDDR5"
  total: "8 GB"
  available_linux: "~6.85 GB"
  bandwidth: "176 GB/s"

storage:
  internal: "500GB-1TB SATA HDD"
  upgradeable: "2.5" SATA SSD"

network:
  ethernet: "Gigabit (Realtek RTL8153)"
  wifi: "802.11 b/g/n (2.4GHz only)"
  bluetooth: "2.1+EDR"
  usb: "USB 3.0 x2, USB 2.0 x1"
```

```yaml
power:
  tdp: "~120W under load"
  idle: "~60W"

cost:
  used_market: "$80-150"

linux_support:
  status: "MATURE"
  kernel: "6.15.4 (latest)"
  gpu_accel: "Full (AMDGPU + Mesa)"
  docker: "Yes"
  kvm: "No (PS4 base)"

limitations:
  - "No AVX2 (limits some optimized code)"
  - "Weak single-thread (~1400 Geekbench 4 SC)"
  - "Thermal throttling common (needs maintenance)"
  - "SATA only (no NVMe)"
```

### PS4 Pro (CUH-70xx series)

```yaml
console:
  model: "PS4 Pro"
  tier: 2
  soc: "AMD Neo"

cpu:
  architecture: "AMD Jaguar x86-64"
  cores: 8
  threads: 8
  clock: "2.13 GHz (OC to 2.6 GHz stable)"
  features: ["AES-NI", "AVX", "SSE4.1", "SSE4.2"]
  # NOTE: NO AVX2

gpu:
  architecture: "AMD GCN 4.0 Polaris"
  compute_units: 36
  tflpos_fp32: 4.20
  memory: "Shared with system"

memory:
  type: "GDDR5"
  total: "8 GB GDDR5 + 1 GB DDR3"
  available_linux: "~6.85 GB GDDR5"
  bandwidth: "218 GB/s"

storage:
  internal: "1TB SATA HDD"
  upgradeable: "2.5" SATA SSD"
```

```yaml
network:
  ethernet: "Gigabit (Realtek RTL8153)"
  wifi: "802.11 a/b/g/n/ac"
  bluetooth: "4.0"
  usb: "USB 3.1 Gen1 x3"

power:
  tdp: "~160W under load"
  idle: "~70W"

cost:
  used_market: "$150-250"

linux_support:
  status: "MATURE"
  kernel: "6.15.4 (latest)"
  gpu_accel: "Full (AMDGPU + Mesa, some accel issues)"
  docker: "Yes"
  kvm: "YES (confirmed working)"

limitations:
  - "No AVX2"
  - "GPU acceleration has minor issues under Linux"
  - "SATA only (no NVMe)"

advantages:
  - "Best cost/performance ratio ($59/TFLOP)"
  - "KVM enables nested virtualization"
  - "WiFi AC for wireless cluster nodes"
  - "Overclockable CPU to 2.6 GHz"
```

### PS5 Base (CFI-10xx/11xx series)

```yaml
console:
  model: "PS5"
  tier: 1
  soc: "AMD Oberon"

cpu:
  architecture: "AMD Zen 2 x86-64"
  cores: 8
  threads: 16
  clock: "3.5 GHz (variable frequency)"
  features: ["AES-NI", "AVX2", "AVX-512? No", "SSE4.2"]
  note: "35% smaller FPU than desktop Zen 2"

gpu:
  architecture: "AMD RDNA 2"
  compute_units: 36
  tflpos_fp32: 10.28
```

```yaml
  ray_tracing: "Yes (hardware)"
  memory: "Shared with system"

memory:
  type: "GDDR6"
  total: "16 GB"
  available_linux: "~12-13 GB"
  bandwidth: "448 GB/s"

storage:
  internal: "825GB custom NVMe SSD"
  expansion: "M.2 NVMe slot (PCIe 4.0 x4)"
  speed_raw: "5.5 GB/s"
  speed_compressed: "8-9 GB/s (Kraken hardware)"

network:
  ethernet: "Gigabit"
  wifi: "WiFi 6 (802.11ax)"
  bluetooth: "5.1"
  usb: "USB 3.1 Gen2 x2, USB 2.0 x2, USB-C"

# CUSTOM I/O COMPLEX (Unique Advantage)
custom_io:
  decompressor: "Kraken/Zlib/Oodle hardware"
  throughput: "8-9 GB/s compressed → raw"
  equivalent_cpu: "~9 Zen 2 cores"
  accessible_linux: "NO (Orbis OS only)"
  accessible_orbis: "YES"

power:
  tdp: "~200W under load"
  idle: "~50W"

cost:
  used_market: "$400-500"

linux_support:
  status: "NEW (April 2026)"
  kernel: "6.8+ (Ubuntu 24.04)"
  gpu_accel: "Full (AMDGPU + Mesa)"
  docker: "Yes"
  kvm: "Expected (Zen 2 has AMD-V)"

limitations:
  - "Jailbreak limited to firmwares 3.xx-4.xx (best) or 7.61 (max)"
  - "Custom I/O decompressor inaccessible from Linux"
  - "Smaller FPU than desktop Zen 2"
  - "Variable frequency may affect benchmarking"

advantages:
  - "Desktop-class CPU (Zen 2 8c/16t)"
```

```
  - "Excellent GPU compute (RDNA2)"
  - "16GB GDDR6 with 448 GB/s bandwidth"
  - "WiFi 6 for high-speed wireless nodes"
  - "Custom NVMe SSD + M.2 expansion"
  - "Hardware decompression via Orbis OS agent"
```

### PS5 Pro

```
console:
  model: "PS5 Pro"
  tier: 1
  soc: "AMD Oberon Plus"

cpu:
  architecture: "AMD Zen 2 x86-64"
  cores: 8
  threads: 16
  clock: "Up to 3.85 GHz"

gpu:
  architecture: "AMD RDNA 2 Extended"
  compute_units: 60
  tflpos_fp32: "33.5 (estimated)"
  ray_tracing: "Yes (enhanced)"

memory:
  type: "GDDR6"
  total: "16 GB GDDR6 + 2 GB DDR5"
  bandwidth: "576 GB/s"

# AI PROCESSING UNIT (Unique)
ai_unit:
  type: "PSSR (PlayStation Spectral Super Resolution)"
  performance: "300 TOPS INT8"
  accessible: "Unknown from Linux"

storage:
  internal: "2TB custom NVMe SSD"
  expansion: "M.2 NVMe slot"

cost:
  used_market: "$550-700 (rare)"

linux_support:
  status: "VERY NEW"
  note: "Userland exploit up to 10.40, kernel exploit limited"
```

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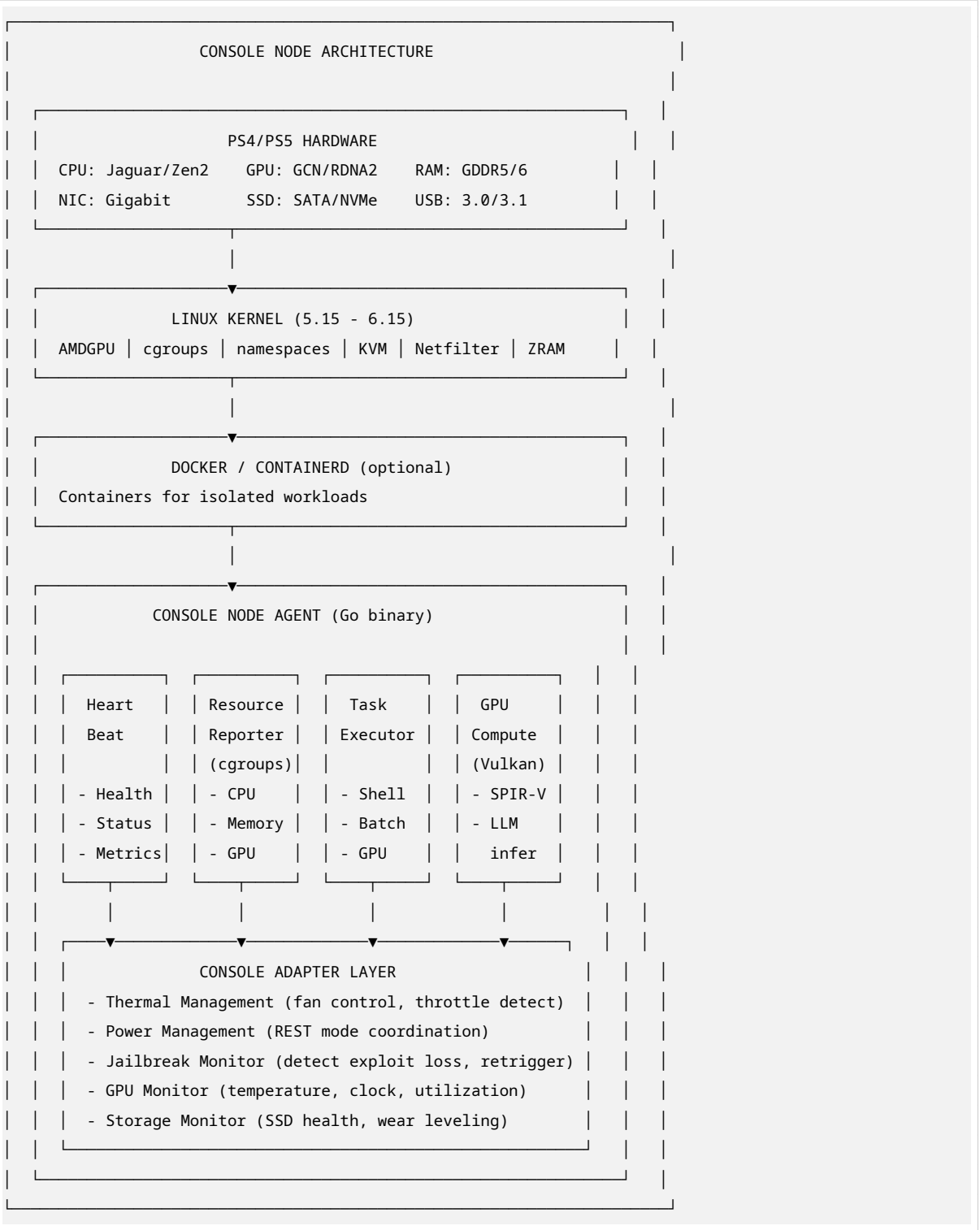
# 3. CONSOLE NODE AGENT ARCHITECTURE

## 3.1 Node Agent Deployment Model

Each console runs a **minimal Linux distribution** (psxitarch, Fedora, or Ubuntu) with our Console Node Agent as a systemd service. The agent provides the same interface as PC node agents but adapts to console-specific constraints.



### 3.2 Console Adapter Layer

The **Console Adapter Layer** is unique to console nodes and handles console-specific concerns:

```
// Console Adapter Interface
package console

type ConsoleAdapter interface {
    // Thermal Management
    GetTemperature() (*ThermalState, error)
    SetFanSpeed(percent int) error
    GetThermalThrottleStatus() (bool, error)

    // Power Management
    GetPowerConsumption() (watts float64, error)
    EnterRestMode() error  // Preserve jailbreak during idle
    WakeFromRestMode() error

    // Jailbreak Management
    IsJailbroken() (bool, error)
    GetJailbreakVersion() (string, error)
    TriggerExploit() error  // Retrigger if lost
    AutoExploitEnabled() (bool, error)

    // GPU Monitoring (console-specific)
    GetGPUClock() (mhz int, error)
    GetGPUUtilization() (percent int, error)
    GetVRAMUsage() (used, total int64, error)
    GetGPUTemperature() (celsius int, error)

    // Storage
    GetSSDHealth() (*SSDHealth, error)
    GetStorageType() StorageType  // SATA SSD / NVMe
}

type ThermalState struct {
    CPUCelsius     int
    GPUCelsius     int
    FanSpeedPercent int
    Throttling     bool
}

type SSDHealth struct {
    WearLevel     int       // 0-100%
    BadBlocks     int
    PowerOnHours  int
    RemainingLife int       // Estimated days
}
```

### 3.3 Console-Specific Resource Reporting

```
// ConsoleNodeResources extends NodeResources with console-specific fields
```

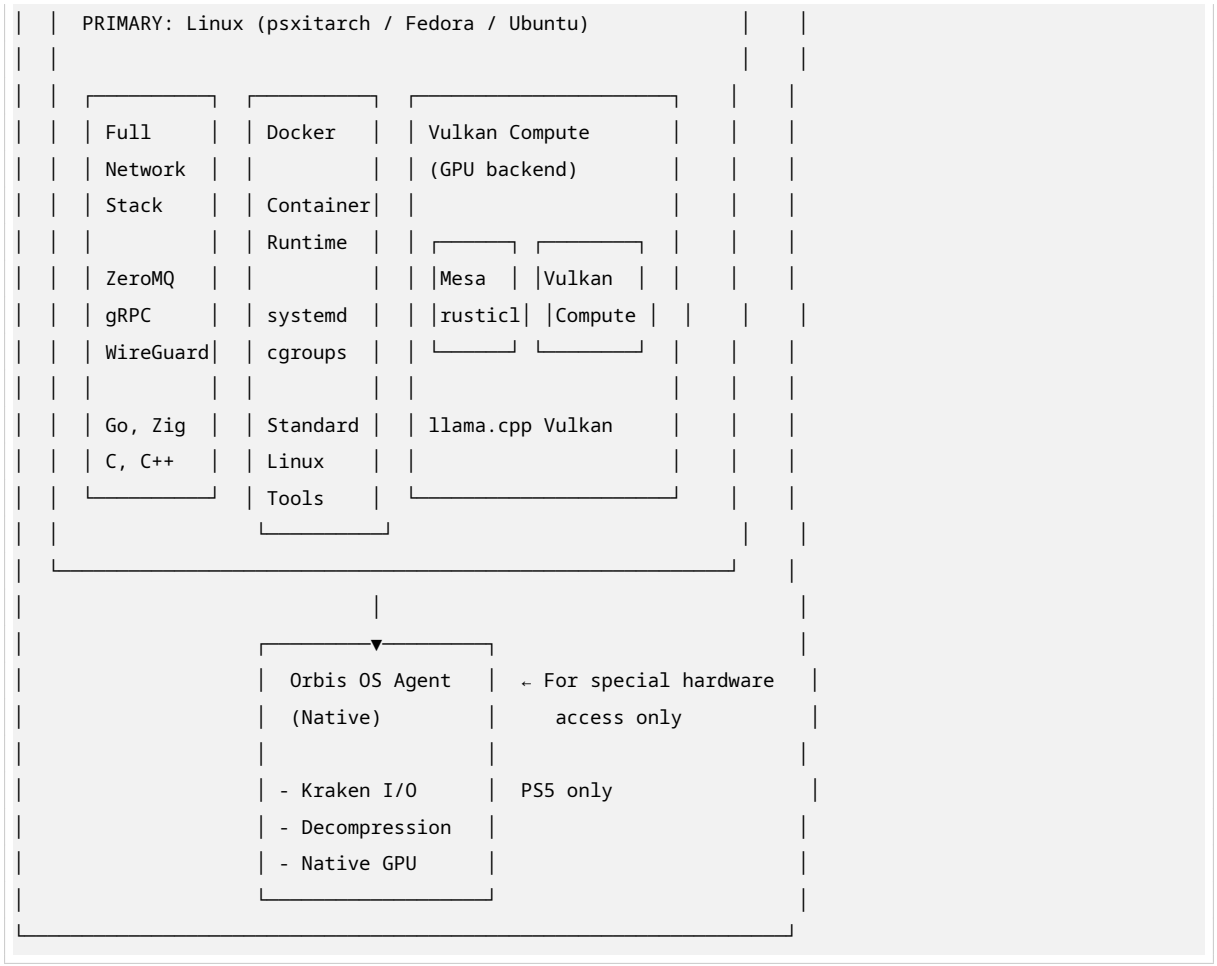
```
type ConsoleNodeResources struct {
    BaseResources  NodeResources      // Standard CPU/Mem/GPU

    ConsoleSpecific struct {
        Model           ConsoleModel  // PS4_FAT, PS4_PRO, PS5, PS5_PRO
        Firmware        string        // e.g., "9.00", "4.51"
        JailbreakType   string        // "GoldHen", "etaHEN", "UMTX2"
        JailbreakVersion string

        Thermal struct {
            CPUCurrentC     int
            GPUCurrentC     int
            FanSpeedPct     int
            Throttling      bool
        }

        Power struct {
            CurrentWatts    float64
            AverageWatts    float64
            PeakWatts       float64
        }

        GPU struct {
            ClockMHz        int
            VRAMUsedBytes   int64
            VRAMTotalBytes  int64
            GPUTemperatureC int
            Throttling      bool
        }

        Storage struct {
            Type            string    // "SATA_SSD", "NVMe_CUSTOM", "NVMe_M2"
            HealthPercent   int
            WearLevel       int
        }

        // PS5 only
        CustomIOAvailable bool       // Kraken decompressor accessible
    }
}
```

---

# 4. OPERATING SYSTEM STRATEGY

## 4.1 Dual-Boot: Linux Primary, Orbis Fallback

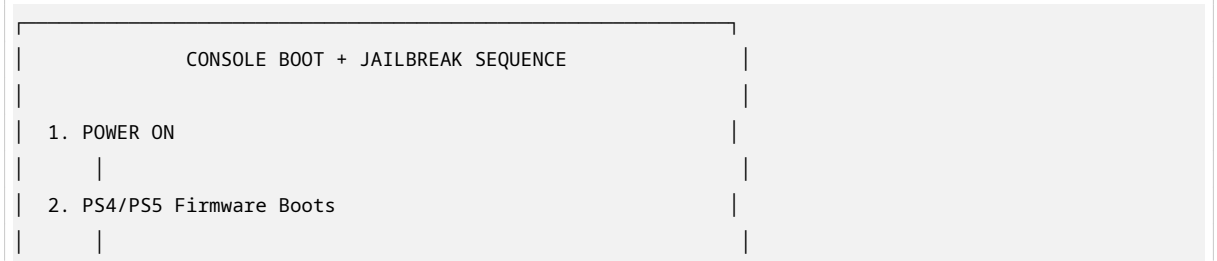
```
┌──────────────────────────────────────────────────────────────┐
│                CONSOLE OS DEPLOYMENT MODEL                   │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐    │
```




### 4.2 Linux Distribution Selection

| Distro | PS4 Support | PS5 Support | Size | Docker | Recommendation |
|---|---|---|---|---|---|
| **psxitarch** | Excellent | None | ~2GB | Yes | **Best for PS4** |
| **Fedora** | Good | Planned | ~4GB | Yes | Best ecosystem |
| **Ubuntu 24.04** | Good | **Best** | ~4GB | Yes | **Best for PS5** |
| **Gentoo** | Good | Unknown | Custom | Yes | Maximum optimization |
| **Arch** | Good | Unknown | ~1GB | Yes | Minimal footprint |

### 4.3 Boot Process with Auto-Exploit



```
│  3. [AUTO-EXPLOIT] ESP32/Luckfox sends USB payload          │
│       │  → UMTX2/GoldHen exploit executes                   │
│       │  → Jailbreak achieved (kernel patch)                │
│       │                                                     │
│  4. GoldHen / etaHEN loads                                  │
│       │  → Plugins initialized                              │
│       │  → FTP server starts (port 2121)                    │
│       │                                                     │
│  5. [LINUX PAYLOAD] ps4-linux / ps5-linux                   │
│       │  → kexec loads Linux kernel from USB/HDD            │
│       │  → Linux takes over hardware                        │
│       │                                                     │
│  6. Linux init (systemd)                                    │
│       │  → Network (dhcpcd/NetworkManager)                  │
│       │  → SSH daemon starts                                │
│       │  → WireGuard mesh interface up                      │
│       │                                                     │
│  7. HelixCluster Node Agent starts                          │
│       │  → Discovers control plane (mDNS/bootstrap)         │
│       │  → Registers with cluster                           │
│       │  → Reports capabilities                             │
│       │  → Ready to accept work                             │
│      ▼                                                      │
│  8. NODE ACTIVE                                             │
└─────────────────────────────────────────────────────────────┘
```

---

## 5. GPU COMPUTE ON CONSOLES

### 5.1 Vulkan Compute: The Universal GPU Backend

**Critical Finding**: Vulkan compute shaders provide a SINGLE API that works across:

- PS4 (GCN 2.0) via RADV or AMDGPU-PRO drivers
- PS4 Pro (GCN 4.0 Polaris) via RADV
- PS5 (RDNA 2) via RADV
- AMD PC GPUs
- Intel Arc GPUs
- NVIDIA GPUs

This means our existing **Vulkan Compute Backend** in the GPU Compute Engine needs NO console-specific modifications.

```
// Example: Universal Vulkan compute shader
// Runs identically on PS4, PS5, and PC GPUs
#version 450

layout(local_size_x = 256, local_size_y = 1) in;
```

```
layout(binding = 0) readonly buffer Input {
    float data[];
} input_buffer;

layout(binding = 1) writeonly buffer Output {
    float data[];
} output_buffer;

layout(binding = 2) readonly uniform Params {
    uint count;
    float scale;
} params;

void main() {
    uint idx = gl_GlobalInvocationID.x;
    if (idx >= params.count) return;
    output_buffer.data[idx] = input_buffer.data[idx] * params.scale;
}
```

### 5.2 GPU Performance Benchmarks

| Workload | PS4 Base | PS4 Pro | PS5 | Desktop RX 6600 |
|---|---|---|---|---|
| **FP32 TFLOPS** | 1.84 | 4.20 | 10.28 | 8.93 |
| **Vulkan Compute (synthetic)** | Baseline | 2.3x | 5.6x | 4.9x |
| **llama.cpp 3B (tok/s)** | ~25 | ~55 | **104** | ~95 |
| **llama.cpp 35B MoE (tok/s)** | ~9 | ~20 | **38** | ~35 |
| **Video Encode (1080p60)** | 1 stream | 2 streams | 4 streams | 3 streams |
| **Matrix Multiply (GFLOPS)** | ~1200 | ~2800 | ~6800 | ~5900 |

*Source: BC-250 benchmarks (PS5-class AMD APU) [dim05]*

### 5.3 llama.cpp AI Inference on Consoles

```
# Build llama.cpp with Vulkan for PS4/PS5
# Runs on any Linux distro on console

cd /opt/llama.cpp
cmake -B build -DLLAMA_VULKAN=ON
make -C build -j$(nproc)

# Run inference using Vulkan GPU backend
./build/bin/llama-cli \
  -m /models/qwen2.5-3b-instruct-q4_k_m.gguf \
  -p "Explain quantum computing:" \
```

```
  -n 256 \
  --gpu-layers 99 \
  --backend vulkan

# For PS5: Much larger models possible
./build/bin/llama-server \
  -m /models/deepseek-v2-lite-q4_k_m.gguf \
  --gpu-layers 99 \
  --ctx-size 8192 \
  --port 8080 \
  --host 0.0.0.0
```

### 5.4 Mesa rusticl for OpenCL on PS4

```
# Enable rusticl for OpenCL on PS4 GCN
export RUSTICL_ENABLE=radeonsi

# OpenCL works for workloads that need it
clinfo  # Shows PS4 GPU as OpenCL device

# Use for hashcat, video encoding, etc.
hashcat -I  # Lists PS4 GPU as OpenCL device
```

---

## 6. NETWORK INTEGRATION

### 6.1 Console Network Stack

```
┌─────────────────────────────────────────────────────────┐
│              CONSOLE NETWORK CONFIGURATION              │
│                                                         │
│  Ethernet: eth0 (Gigabit, ~940 Mbps)                    │
│  WiFi:     wlan0 (PS4: 802.11n, PS5: 802.11ax WiFi 6)   │
│                                                         │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐   │
│  │  WireGuard   │  │   ZeroMQ     │  │   HTTP/gRPC  │   │
│  │  Mesh Peer   │  │  Client      │  │   Client     │   │
│  │              │  │              │  │  (lightweight)│   │
│  │ UDP/51820    │  │ TCP/5555+    │  │ TCP/8443     │   │
│  └──────────────┘  └──────────────┘  └──────────────┘   │
│                                                         │
│  PS4: Uses ZeroMQ for control (lightweight)             │
│  PS5: Can use gRPC directly (Zen 2 powerful enough)     │
│  Both: WireGuard for encrypted mesh                     │
└─────────────────────────────────────────────────────────┘
```

### 6.2 Protocol Selection by Console

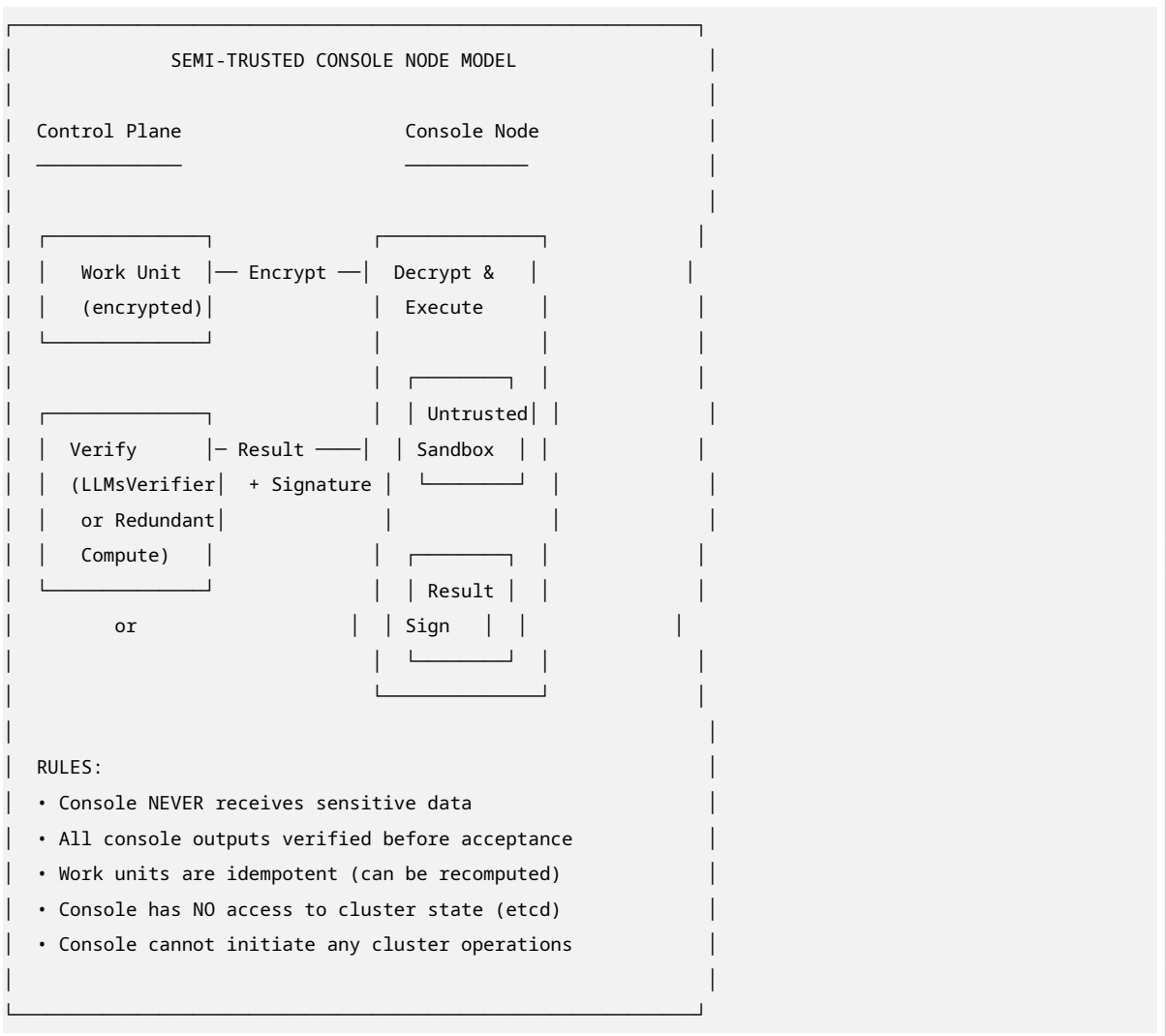
| Protocol | PS4 | PS5 | Notes |
|---|---|---|---|
| **WireGuard** | | | Kernel module, ~800 Mbps |
| **ZeroMQ** | (primary) | | Lightweight, no deps |
| **gRPC** | (too heavy) | | Zen 2 handles it |
| **NATS** | (limited) | | JetStream may struggle on PS4 |
| **Arrow Flight** | | | PS4 lacks AVX2 for efficient processing |
| **SSH** | | | Native Linux |
| **WebSocket** | | | For session I/O |

---

## 7. SECURITY MODEL FOR CONSOLE NODES

### 7.1 Semi-Trusted Node Architecture

Console nodes operate at **TRUST_LEVEL = SEMI**. They receive work units but never sensitive data.



### 7.2 Security Measures

| Layer | Measure | Rationale |
|---|---|---|
| **Data** | Work units encrypted with node-specific key | Console kernel is compromised |
| **Verification** | All results verified by LLMsVerifier or redundant compute | Detect tampering |
| **Network** | WireGuard only, no direct access to cluster services | Isolate console network |
| **Access** | Console nodes read-only to cluster state | Cannot modify anything |
| **Workload** | Only idempotent, non-sensitive tasks | Safe to recompute if needed |
| **Monitoring** | Anomaly detection on console outputs | Detect compromised nodes |

---

## 8. CONSOLE WORKLOAD PROFILES

### 8.1 Optimal Workloads for Console Nodes

| Workload | PS4 | PS4 Pro | PS5 | Notes |
|---|---|---|---|---|
| **AI Inference (llama.cpp)** | Small models | Medium models | **All models** | Vulkan backend |
| **Video Transcode** | 1x 1080p | 2x 1080p | 4x 1080p | Vulkan compute shaders |
| **AOSP Build (distributed)** | Limited | Good | **Excellent** | CPU-bound compilation |
| **Hash/Crypto Cracking** | Adequate | Good | **Excellent** | GPU-parallel |
| **Data Decompression** | Software | Software | **Hardware (Kraken)** | PS5 Orbis agent only |
| **Image Processing** | Slow | Adequate | **Fast** | GPU compute |
| **Machine Learning Training** | No | Small models | Medium models | Limited VRAM |
| **Ray Tracing** | No | No | **Yes** | Hardware RT on PS5 |
| **Web Serving** | Yes | Yes | Yes | Light HTTP/WS |
| **Cache/Storage Node** | Yes (SATA) | Yes (SATA) | **Yes (NVMe)** | ZFS/bcache |

### 8.2 Workloads to AVOID on Consoles

| Workload | Reason |
|---|---|
| Double-precision scientific computing | No FP64 hardware on GCN/RDNA |
| Memory-intensive >8GB (PS4) / >12GB (PS5) | Limited RAM |

Table 7 – continued

| Workload | Reason |
|---|---|
| Single-threaded latency-sensitive tasks | Weak Jaguar cores on PS4 |
| Trusted/attested computation | Kernel is compromised |
| Persistent state storage | Nodes are disposable |
| Work requiring AVX2 | PS4 lacks AVX2 |

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# 9. SETUP WIZARD: CONSOLE PROVISIONING

## 9.1 Console Node Setup Flow

```
$ htmux cluster add-console

[1/8] Detecting console via network scan...
       Found: PS4 Pro (CUH-7016B) at 192.168.1.45
       Firmware: 9.00  (exploitable)

[2/8] Preparing jailbreak payload...
       Loading GoldHen v2.4b via USB auto-exploit...
       [██████████] 100% Jailbreak successful

[3/8] Installing Linux payload...
       Loading ps4-linux (kernel 6.15.4)...
       Booting Linux from USB...
       [██████████] 100% Linux running

[4/8] Installing HelixCluster Console Agent...
       Downloading: console-agent-linux-amd64
       Installing systemd service...
       [██████████] 100% Agent installed

[5/8] Configuring network...
       WireGuard keypair generated
       Mesh tunnel established to control plane
       IP assigned: 100.64.2.15

[6/8] Running hardware capability scan...
       CPU: AMD Jaguar x8 @ 2.13 GHz
       GPU: AMD GCN 4.0, 36 CUs, 4.20 TFLOPS
       RAM: 8 GB GDDR5 @ 218 GB/s
       Storage: 500GB SATA SSD
       Network: Gigabit Ethernet

[7/8] Running GPU compute test...
       Vulkan compute: PASS (2.1 TFLOPS sustained)
       llama.cpp 3B: 55 tok/s

[8/8] Registering with cluster...
```

```
      Node ID: c4f8e2d1-...
      Trust Level: SEMI
      Tier: 2 (Standard)
      [██████████] 100% Console node registered!

Console node 'PS4-Pro-LivingRoom' is now part of your cluster.
Available for: GPU compute, batch jobs, AI inference, video transcode
Type: htmux cluster status to see all nodes.
```

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## 10. PHASE 2 IMPLEMENTATION PLAN

### 10.1 New Tasks for Console Integration

| Phase | Task | Description | Hours | Priority |
|---|---|---|---|---|
| **0** | C-0.1 | Console Agent scaffolding (Go, Linux-only) | 8 | P0 |
| **0** | C-0.2 | Console Adapter interface definition | 4 | P0 |
| **0** | C-0.3 | Thermal/power monitoring library (PS4/PS5) | 8 | P0 |
| **0** | C-0.4 | Jailbreak detection and auto-trigger library | 8 | P0 |
| **0** | C-0.5 | Auto-exploit hardware firmware (ESP32) | 16 | P1 |
| **0** | C-0.6 | Console capability scanner | 4 | P0 |
| **0** | C-0.7 | PS5 I/O Agent (Orbis OS native, Kraken) | 16 | P2 |
| **1** | C-1.1 | Console node registration with SEMI trust | 4 | P0 |
| **1** | C-1.2 | Console-specific heartbeat (thermal, power) | 4 | P0 |
| **1** | C-1.3 | WireGuard on PS4/PS5 (kernel module) | 4 | P0 |
| **1** | C-1.4 | ZeroMQ client for PS4 (lightweight protocol) | 4 | P0 |
| **1** | C-1.5 | gRPC client for PS5 (full protocol) | 4 | P0 |
| **2** | C-2.1 | Vulkan GPU backend testing on PS4/PS5 | 8 | P0 |
| **2** | C-2.2 | llama.cpp Vulkan integration for console AI | 8 | P0 |
| **2** | C-2.3 | Console-specific ClassAds (thermal, GPU throttling) | 4 | P0 |
| **2** | C-2.4 | Workload suitability scheduler plugin | 8 | P0 |

Table 8 – continued

| Phase | Task | Description | Hours | Priority |
|---|---|---|---|---|
| **3** | C-3.1 | Console session backend (minimal PTY) | 8 | P1 |
| **4** | C-4.1 | AOSP build worker on PS4 (distcc volunteer) | 8 | P1 |
| **4** | C-4.2 | AOSP build worker on PS5 (distcc + GPU) | 8 | P1 |
| **5** | C-5.1 | Console AI inference agent (llama.cpp) | 8 | P1 |
| **7** | C-7.1 | Console chaos tests (power loss, thermal) | 8 | P0 |
| **7** | C-7.2 | Console verification testing (SEMI model) | 8 | P0 |
| **8** | C-8.1 | Console setup wizard integration | 8 | P0 |
| **8** | C-8.2 | Auto-exploit hardware provisioning | 8 | P1 |

**Total Phase 2 Additional Tasks: 24 tasks, ~176 hours (~4.5 weeks)**

### 10.2 Integration Points with Existing Architecture

| Existing Component | Console Integration |
|---|---|
| **Node Discovery** | Console nodes register with CONSOLE tier tag |
| **Resource Scheduler** | ConsoleClassAds plugin for thermal/GPU throttling awareness |
| **Session Manager** | Console nodes get minimal PTY sessions (no migration) |
| **GPU Compute Engine** | Vulkan backend (already universal, no changes needed) |
| **Health Monitor** | Console-specific thermal/power metrics |
| **LLM Brain** | Console AI inference pool for parallel agents |
| **Build Service** | Console distcc workers for AOSP compilation |
| **Security Manager** | SEMI trust level, encrypted work units, output verification |
| **Backup Service** | Console nodes excluded from backup (stateless) |

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## 11. GAPS FILLED BY CONSOLE NODES

| Gap | How Consoles Fill It | Impact |
|---|---|---|
| **GPU Compute Cost** | 1/2 the cost per TFLOP vs PC GPUs | 50% savings on GPU pool |
| **GDDR5/ GDDR6 Bandwidth** | 176-448 GB/s vs DDR4' s 25-50 GB/s | 7-18x memory bandwidth for GPU workloads |
| **Decompression Acceleration** | PS5 Kraken: 8-9 GB/s hardware decompression | 9 CPU cores freed per PS5 |
| **Disposable Compute** | At $80-250, replace not repair | Simplified ops, no maintenance |
| **Community Scaling** | Users donate idle consoles | Elastic capacity without purchase |
| **Power Efficiency** | 0.027-0.051 TFLOPS/Watt | Lower power bills |
| **GPU Shortage Hedge** | Alternative to unobtainable PC GPUs | Continued scaling during shortages |